# Math for CS & AI: Homework 10

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(Adapted from earlier versions of this course taught by Andrew Yao)

**Due Date: May 30, 2026**

## Study Tips

Some students have asked for additional exercises to practice their problem-solving skills. These exercises are not required, but if you are interested, you may find some useful problems in the following textbook:

- *Foundations of Data Science*, by Avrim Blum, John Hopcroft, and Ravindran Kannan
  - Chapter 2: High-Dimensional Space
- *All of Statistics: A Concise Course in Statistical Inference*, by Larry Wasserman
  - Chapter 2: Random Variables
  - Chapter 3: Expectation
  - Chapter 4: Inequalities

You may also want to practice your calculus skills to truly master continuous probability. Check any standard calculus textbook for exercises on integration and differentiation.

**Note:** Skip the problems that require more knowledge than what is covered in this course.

## Problem Set

- Exercises 5.9–5.18 from Chapter 5 of the lecture notes:
  https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

## Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.

- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

# Additional Hints

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.
- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.




Think twice before scrolling down!

# Hints

## Problem 5.9

- (a) If $\mathrm{Cov}(X, Y) = 0$, then $(X, Y) = \boldsymbol{A}\boldsymbol{z}$ for some $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ and some diagonal matrix $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$.
- (b) Just compute the covariance.

## Problem 5.10

- (b) $X = \frac{1}{2}Y + \frac{1}{2}(X - Z)$. What is the relationship between $Y$ and $X - Z$ again?

## Problem 5.11

- $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{A}^\top \boldsymbol{x} = \|\boldsymbol{A}\boldsymbol{x}\|_2^2$.
- $(\boldsymbol{Q}\boldsymbol{\Lambda}^{1/2}\boldsymbol{Q}^t op)^2 = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^t op$.

## Problem 5.12

- If $|u_i| \neq |\ell_i|$, then the volume of $H$ is not optimal.
- $\sum_i u_i^2 \leq 1$.

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## Problem 5.13

- The rate function is $\frac{1}{2}(\epsilon - \log(1+\epsilon)) = \Theta(\epsilon)$ when $\epsilon \to +\infty$.
- $\frac{\log(1+\epsilon)}{\epsilon}$ is decreasing in $\epsilon$.
- When $\epsilon \geq 1$, $\|\boldsymbol{x}\|_2 - 1 \geq \epsilon \iff \|\boldsymbol{x}\|_2^2 \geq 1 + 2\epsilon + \epsilon^2$.

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## Problem 5.14

- Write $Z_1, Z_2$ in polar coordinates $(r, \theta)$.
- $\theta$ is uniform on $[0, 2\pi]$.
- $r^2$ follows an exponential distribution.

## Problem 5.15

- Read the lecture notes on JL lemma.
- Do random projection with parameter $\epsilon_0$.
- The diameter $D$ becomes at least $(1-\epsilon_0)^{1/2}D$ after projection.
- For any pair of points $x_i, x_j$ with distance $< (1-\epsilon)D$, the distance becomes $< \frac{(1-\epsilon)D}{(1-\epsilon_0)^{1/2}}$.
- How to choose $\epsilon_0$?
- Remeber to describe how to generate your Gaussian random matrix.

## Problem 5.16

- Complement event.

## Problem 5.17

- (a) $\mathbb{E}d_{\mathrm{TV}}(\hat{P}^{(n)}, P) = \frac{1}{2}\sum_x \mathbb{E}|p_n(x) - p(x)|$. Bound each term by $\mathbb{E}[(p_n(x) - p(x))^2]$.
- (b) Do you still remember Dr. McDiarmid?

## Problem 5.18

- (a) Don't integrate Guassian by your self! Use the formula for the Gaussian MGF.
- (b) Let $I := \int_{-\infty}^{+\infty} |f(x)| \, \mathrm{d}x$. Then

$$\int_{-\infty}^{+\infty} |\mathbb{E}[f(x+Z)]| \, \mathrm{d}x \leq \mathbb{E}\left[\int_{-\infty}^{+\infty} |f(x+Z)| \, \mathrm{d}x\right] = \mathbb{E}[\,?\,].$$

- (c) You may see this integral:

$$\int_{\mathbb{R}} \underbrace{\int_{\mathbb{R}} f(x_0) e^{i\omega x_0} e^{-\frac{1}{2}\sigma^2\omega^2} e^{-i\omega x} \, \mathrm{d}\omega}_{?} \, \mathrm{d}x_0.$$

- (d) As $\sigma \to 0$, $\mathcal{F}[f_\sigma] = \mathcal{F}[f] \cdot (\text{something}) \to \mathcal{F}[f]$. So $f_\sigma = \mathcal{F}^{-1}[\mathcal{F}[f_\sigma]] \to \mathcal{F}^{-1}[\mathcal{F}[f]]$.
- What is $\lim_{\sigma\to 0} f_\sigma(x)$? For any $\delta > 0$, let $(x, x+\epsilon)$ be the interval so that $|f(x) - f_+(y)| < \delta$ for all $y \in (x, x+\epsilon)$. Then as $\sigma \to 0$,

$$\mathbb{E}[f(x+Z)\mathbb{1}_{[Z>0]}] = \underbrace{\mathbb{E}[f(x+Z)\mathbb{1}_{[Z\in(0,\epsilon)]}]}_{\to f(x)\pm\delta} + \underbrace{\mathbb{E}[f(x+Z)\mathbb{1}_{[Z\geq\epsilon]}]}_{\leq I\cdot\Pr[Z\geq\epsilon]}.$$

- (e) $M_X(it)$ is just the Fourier transform of $f_X(x)$.